Research Designs

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1. Introduction

Research design provides a framework for the collection and analysis of data, shaping how research questions are addressed and how conclusions are drawn. The process begins with defining a clear research question**,** which serves as the foundation for a study. Once the research question is established, researchers develop a research design; the overall plan that determines how the question will be addressed. Only after the design is finalized do researchers begin to collect and analyze data**,** guided by the decisions made in the design phase.

Research design is about the study's overall structure. It guides which methods you use and helps ensure the study matches its objective. Importantly, research design is distinct from research methods: while methods are specific techniques for collecting data (e.g., surveys or interviews), research design reflects broader choices about the overall approach, priorities, and organization of the research.

A helpful analogy is the construction of a building (cf. de Vaus 2001). The research design is similar to deciding what type of building we want to construct—whether it will be a skyscraper, a house, or a factory. This decision informs everything that follows. It affects which methods you use and what data you collect, etc. The primary function of a research design is to ensure that the research question can be answered, given the methods, data, and overall structure of the study. A well-thought-out design aligns these elements to produce credible and meaningful results.

Bell et al. (2022) argue that different research designs, assign different levels of importance to the following **four dimensions:**

- "expressing causal connections between variables;
- generalizing to larger groups of individuals than those actually forming part of the investigation;
- understanding behaviour and the meaning of that behaviour in its specific social context;
- having a temporal (i.e., over time) appreciation of social phenomena and their interconnections." (Bell et al. 2022, p. 40).

In this chapter, we explore three prominent research designs—**cross-sectional, longitudinal, and case studies**—each reflecting distinct priorities regarding these dimensions (see Bergenholtz et al. 2026a for information on **experiments**, a fourth research design). Understanding their strengths and weaknesses will enable you to critically assess research studies and do an informed selection of designs in future work.

2. Cross-sectional design

A cross-sectional design captures data at a single point in time, providing a snapshot of relationships between variables. This design is particularly effective for exploring **generalizations to larger groups of individuals** (cf. the bullet points nr. 2 in section 1 just above) and for identifying correlations between variables. However, it is less useful for understanding behavior in

specific contexts or capturing temporal changes over time. For instance, instead of asking why a specific individual chose a particular education or job, a cross-sectional design focuses on identifying the **average** relation between different types of education and salary - while potentially statistically controlling for other confounding factors (see this link for an explainer of how statistical controls work. See also Bergenholtz 2026e that briefly explains the intuition of controls).

Illustrating a cross-sectional design

To visualize cross-sectional research, imagine an Excel sheet where each row represents an individual (e.g., employees or students), and each column represents a variable, such as education level, salary, years of experience, job satisfaction, and industry type. Each cell contains one specific data point for each variable (e.g. education and salary) for an individual, creating an extensive dataset for analysis.

Respondent	Education	Salary / Month	Years of experience	Job satisfaction (1-5)	Industry type
1	BA	42.000	3	3	IT
2	MA	52.000	6	4	Finance
3	College	32.000	5	2	Manufacturing
4	BA	48.000	2	5	Finance
5	BA	38.000	8	5	Manufacturing
6	MA	60.000	10	3	IT
7	PhD	65.000	5	4	University
…	…	…	…	…	…

Table 1: Illustrating cross-sectional data.

A cross-sectional design still gives you one data point per participant. Data collection can take time, but you end up with a single value for each person. For example, imagine a researcher who collects data on how much people exercise over a period of a month. The researcher asks people once a week how much they exercised. But in the analysis, the researcher keeps only one value per participant. For example, they use the participant's average across the weeks. This approach would be cross-sectional, since only one data point is available, and the focus is on summarizing relationships across variables and participants, not on investigating changes over time within that month. Also note that one does not have to collect the data oneself; one can use data others have collected.

Strengths and weaknesses

A well-designed cross-sectional study can **suggest a possible causal link**. This is more plausible when the sample is large and the study controls for important confounders. However, there is always a risk that any **observed relationships between variables might be influenced by unmeasured factors** (omitted variable bias, see Bergenholtz 2026e). For example, a study might find a correlation between exercise and mood. The researcher can control for factors like weight, diet, stress, and socioeconomic status. But we still cannot be sure all confounders are measured well, or that no important confounder is missing (e.g., genetic factors). In contrast, in an RCT, participants are randomly assigned to treatment or control. This makes the groups similar on average. As a result, confounders like weight, diet, stress, and socioeconomic status should be evenly balanced across groups - although randomization might fail if sample sizes are too small, cf. https://carstenbergenholtz.github.io/Phil-of-Science-tools/randomization_sample_size.html.

Generally, cross-sectional designs prioritize **external validity** (the ability to generalize findings to larger populations) over **internal validity** (confidence in finding causal relationships in this particular dataset). This approach also reflects specific assumptions about how knowledge is created in social science. By focusing on patterns across many cases rather than deeply analyzing specific instances (as a case study, see section 4), cross-sectional designs emphasize identifying broad, generalizable trends. This aligns with the goal of discovering consistent relationships across populations rather than understanding the unique details of individual cases.

3. Longitudinal design

A longitudinal design involves collecting data from the same entities (e.g. individuals, groups or organizations) at multiple points in time. This approach allows researchers to observe changes and trends, offering insights into how variables develop or relate to each other over time. Unlike cross-sectional studies, which capture a single moment in time, longitudinal studies follow individuals, groups, or organizations over an extended period, tracking their progression or change. There are two main types of longitudinal studies:

- **Panel studies**: These involve repeatedly collecting data from the same participants over time. The key feature is that the same individuals are measured repeatedly. This makes it possible to track individual-level change. **Example**: Following the same 100 employees every quarter to monitor their exercise habits and mood over a year, even if some participants eventually leave the study.

- **Cohort studies**: A type of longitudinal design that tracks a specific sub-group (a cohort) sharing a defining characteristic. Data are collected in repeated survey waves at intervals through time to see how the cohort as a group develops. Unlike panel studies, the same individuals are not necessarily surveyed in each wave; rather, different samples are often drawn from the same cohort. **Example**: Tracking "employees hired in 2020" by surveying a different random sample of this group each quarter to understand their general career progression patterns, without necessarily reaching the same specific individuals each time.

Longitudinal research primarily addresses two dimensions from the introduction: **expressing causal connections between variables** and **having a temporal (over time) appreciation of phenomena**. Longitudinal studies show the order of events over time. This can give stronger evidence about causality than cross-sectional studies. For example, if increase in exercise consistently happens before improvements in mood over a period of time, this suggests a directional relationship.

Respondent	Time Point	Exercise Frequency (Days/Week)	Mood Rating (1-5)
1	T1	2	3
1	T2	4	4
1	T3	5	5
2	T1	0	2
2	T2	1	3
2	T3	2	4
3	T1	3	4
3	T2	3	4
3	T3	4	5
4	T1	1	2
4	T2	2	3
4	T3	3	4
...	...	...	...

Table 2: Illustrating a longitudinal design. Note there are multiple data points for each individual.

While this suggests a relationship, the evidence is not conclusive. Maybe measurements of mood are somewhat delayed. Or, maybe an omitted factor—such as participation in a wellness program that encourages both exercise and stress management—could be influencing both exercise and mood (cf. Isager 2023). Without accounting for such omitted variables or measurement challenges, the study might not correctly assess the direct link between exercise and mood.

Strengths and weaknesses

Compared to cross-sectional designs, longitudinal designs often prioritize **internal validity** by tracking changes over time and establishing temporal sequences. By following such changes, there is an opportunity to measure whether a change in the independent variable happens before a change in the dependent variable. This is something a cross-sectional study cannot ensure. However, longitudinal studies typically focus on specific populations and might involve smaller samples, which can limit **external validity**, making generalization to broader populations more challenging. Furthermore, the risk of omitted variable bias is still in play, as in cross-sectional studies. Just because one has data over time (thus knows what happened first), does not mean that one has eliminated the risk of overlooking an important variable. A few additional methodological challenges deserve attention in longitudinal research:

- Time-Related Effects: When conducting research over time, several factors can affect results independently of what is being studied. Natural developments, changing social contexts and people dropping out of a study can create biases that are difficult to identify. For example**:** Imagine one is studying a situation where employees are getting specific on-the-job training, and one is interested in seeing the effects of this over time. If employees demonstrate an

improved job performance over time, this might be due to general on-the-job experience one would have gotten anyway, rather than the specific training programs being studied (the natural development). Or, shifting cultural norms could affect how survey questions are answered over time, meaning what looks like a change in performance, could be due to how performance ratings are created and perceived by employees and managers (changing social contexts). Finally, employees who quit might be systematically different from those who remain. Hence, if over time one sees a change in a relevant variable, this might simply be due to some people (low performers for example) dropping out (also called attrition bias). Researchers must account for these time-related factors and participation patterns to be sure to make valid conclusions.

Summing up
Longitudinal research is centered on the idea that people, behaviors, and societies are dynamic—they change over time. This approach emphasizes the importance of assessing these changes, not only to understand behaviors better but also to improve our ability to address causality. By examining patterns and developments across time, longitudinal studies provide insights into both the mechanisms of change and the factors that drive change.

4. Case study design

A case study design focuses on an in-depth exploration of a specific, bounded system, such as an organization, event, individual, or process. Rather than attempting to understand averages or general trends across populations, case studies aim to uncover the nuanced dynamics and unique characteristics of the subject under investigation. Referring back to the four bullet points in the introduction, this design **aims to understand behavior and the meaning of that behavior in its specific social context**. By diving deeply into a particular case, researchers create insights into the particular factors and interactions that shape what happens in the case context.

Illustrating a case study design
Imagine a case study examining why implementing chatbots in a customer service organization might be challenging. The researcher could use several methods—such as observing chatbot usage during customer interactions, conducting interviews with employees and managers, distributing surveys to gauge employee attitudes, and analyzing company performance data. The aim would be to investigate how various factors (e.g. organizational culture, employee skills, and existing workflows) shape the adoption of the chatbot system.

To illustrate the nature of a case study, one can contrast it with the cross-sectional design: The **kinds of research questions asked in cross-sectional and case study designs are very different** in focus and aim. In a cross-sectional design, questions typically aim to identify general patterns or correlations across a population at a single point in time. For example: “What is the relationship between job satisfaction and productivity across employees in a customer service industry?” The emphasis is on general and statistical patterns.

In contrast, a case study design focuses on deep, context-specific questions that seek to understand the unique interactions and processes within a single or small set of cases. For instance: “How do frontline employees and managers in a specific customer service company make sense of introducing chatbots, and how do these interpretations show up in day-to-day work practices?” These questions emphasize depth, context, and the meaning of behaviors or events, rather than broad patterns across many data points.

Multiple case studies: In some instances, researchers might adopt a multiple case study design to compare and contrast findings across different contexts (Eisenhardt 1989). Cases are typically selected not for statistical representativeness but for their ability to highlight theoretical differences or similarities (Flyvbjerg 2006). For example, when studying chatbot implementation, organizations might be chosen to represent different levels of technological maturity or different organizational cultures. Single case studies provide an opportunity to focus deeply on a particular context, offering rich and nuanced insights into the unique dynamics of that case. In contrast, multiple case studies (e.g., examining 4 or 8 cases) focus less on the individual uniqueness of each context and more on identifying variations or similarities across cases (Dyer and Wilkins 1991). This approach is arguably somewhat related to the aim of cross-sectional designs, as it emphasizes comparative analysis and generalization across contexts while still using the depth of qualitative insights from each case. By comparing these cases, researchers can identify patterns, refine theories, and draw broader conclusions, thus trying to capture both the cases' depth and broader applicability (Eisenhardt 1989).

Longitudinal: Case studies can also be longitudinal, tracking the process of adoption and integration over time. For example, the researcher might study the organization during the planning, initial implementation, and post-implementation phases to understand how challenges change and how the organization responds.

Strengths and weaknesses
Case studies do well in providing rich, detailed insights into specific phenomena, especially when the research question revolves around "how" or "why" something happens. They are particularly valuable for understanding complex, context-specific interactions that might be missed in broader quantitative, e.g. cross-sectional, studies. Case study research is usually related to a kind of explanation that focuses on understanding as a primary concern, rather than causality (see Bergenholtz 2026f).

However, their findings often have limited **external validity**, as the depth and specificity of the analysis may not transfer to other contexts. Case studies can also be resource-intensive and subject to researcher bias, as the interpretation of data often depends on the researcher's perspective and approach.

Summing up
This design reflects an assumption that knowledge is created through deep, context-specific exploration. By focusing on the unique details of a case, researchers aim to understand the phenomenon itself and the interactions that shape it within its specific case context. Case studies are not designed to uncover general relations, but they can generate valuable insights and inform broader theories, policies, or practices.

5. Comparing research designs

How to identify different research designs (e.g. in an exam situation)
To determine the appropriate research design, or to identify which design is being used in a study, consider the focus and structure of the research. If the study collects data from individuals at a single point in time to investigate patterns or correlations—such as surveying employees' job satisfaction and performance in one week—it is a **cross-sectional design**. If the study instead collects data from the same individuals at multiple points over time—such as measuring employees'

job satisfaction before, during, and after implementing a new policy—it is a **longitudinal design**, which emphasizes observing change or development. When the study focuses on a specific case (or few cases), such as understanding why a particular company struggled to adopt a new technology, and integrates methods like interviews, observations, surveys, or other perspectives, it is a **case study design**, prioritizing contextual understanding. If the study involves manipulating a variable—such as giving randomly selected employees access to a training program while not giving it to other employees to measure differences in performance—it is an **experimental design**, which focuses on isolating causal relationships through control and randomization (see Bergenholtz 2026a).

Combining research designs

Research designs aren't always used in isolation. Studies can combine multiple designs to take advantage of their complementary strengths. For instance, a researcher might use cross-sectional surveys across multiple organizations to identify average patterns while conducting in-depth case studies of selected companies to understand the mechanisms that drive those patterns. One could also do a longitudinal field-experiment that even includes a case study where one observes how the treatment intervention shapes human behavior.

On generalization (external validity)

Generalizability differs significantly across research designs and is closely tied to their assumptions about causality and the nature of their findings.

Cross-sectional and experimental designs often aim for **statistical generalizability**, arguing that findings from a sample (if representative) can be applied to a larger population. These designs operate on the assumption that causality is somewhat stable and predictable. For example, an experiment might test whether offering bonuses increases productivity and, if successful, claim that this relationship holds across similar contexts. Repeated replication of such studies (redoing them in either the same or different contexts) further enhances the reliability of their conclusions.

In contrast, case studies aim for **theoretical generalizability** (or transferability in a constructivist perspective). This approach asks whether insights from a particular context can inform understanding in similar settings. For instance, a case study investigating the introduction of a bonus scheme in a single company might focus on how factors like organizational culture or employee expectations are important for how the scheme is implemented by managers and perceived by employees. While these findings are not statistically generalizable, they can offer valuable lessons for other organizations with similar characteristics.

Basic philosophy of science assumptions

These differences in generalizability reflect deeper **epistemological and ontological** distinctions (see Guba 1990, Bergenholtz 2026f, curriculum in later lessons). Cross-sectional and experimental designs are typically associated with **(post)positivist approaches**, prioritizing objectivity and the search for existing, underlying, general laws of business practices and outcomes. Case studies, on the other hand, can either rely on a post-positivistic approach or reflect **constructivist perspectives.** The latter emphasizes the richness and particularity of individual contexts and the idea that researchers actively construct an understanding of a context. These philosophical underpinnings influence not only how research is conducted but also how its findings are interpreted and applied.

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Bergenholtz, C. (2026d). Research Designs. Course material, Aarhus University, Brightspace.

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